→ Integration is reverse process of differentiation.

Standard Integrals :

1) $\int x^n \cdot dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

2) $\int \frac{1}{x} \cdot dx = \ln|x| + C$

3) $\int e^x \cdot dx = e^x + C$

4) $\int a^x \cdot dx = \frac{a^x}{\ln(a)} + C$

5) $\int \sin x \cdot dx = -\cos x + C$

6) $\int \cos x \cdot dx = \sin x + C$

7) $\int \sec^2 x \cdot dx = \tan x + C$

8) $\int \operatorname{cosec}^2 x \cdot dx = -\cot x + C$

9) $\int \sec x \cdot \tan x \cdot dx = \sec x + C$

10) $\int \operatorname{cosec} x \cdot \cot x \cdot dx = -\operatorname{cosec} x + C$

11) $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$ [If (−) sign present then $\cos^{-1} x$]

12) $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$ [If (−) sign present then $\cot^{-1} x$]

13) $\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1} x + C$ [If (−) sign present then $\operatorname{cosec}^{-1} x$]

Theorems of Integration :→

1) Constant in product with function can be taken out of integration.

$$\int k \cdot f(x) \cdot dx = k \int f(x) \cdot dx$$

2) Sign of integration can be distributed over addition and subtraction.

$$\int [f(x) \pm g(x) \pm h(x)]\, dx = \int f(x) \cdot dx \pm \int g(x)\, dx \pm \int h(x) \cdot dx$$

3) If coeff. of 'x' is not 1, then entire integral is divided by coeff. of x.

$$\int f(ax \pm b) \cdot dx = \frac{1}{a} F(ax \pm b)$$

Ques If $f'(x) = \tan^{-1}(\sec x + \tan x)$, $-\pi/2 < x < \pi/2$, and $f(0) = 0$, then $f(1)$ is equal to ____.

Sol'n $f'(x) = \tan^{-1}\left(\frac{1}{\cos x} + \frac{\sin x}{\cos x}\right) = \left(\frac{1+\sin x}{\cos x}\right)$

$= \tan^{-1}\left(\frac{(\cos x/2 + \sin x/2)^2}{\cos^2\frac{x}{2} - \sin^2\frac{x}{2}}\right)$

Indefinite Integration-1

$f'(x) = \tan^{-1}\left(\frac{\cos\frac{x}{2} + \sin\frac{x}{2}}{\cos\frac{x}{2} - \sin\frac{x}{2}}\right)$

$f'(x) = \tan^{-1}\left(\frac{1+\tan x/2}{1-\tan x/2}\right) = \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right]$

$\int f'(x) \cdot dx = \int \frac{\pi}{4} \cdot dx + \int \frac{x}{2} \cdot dx$

$f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + C$

$\because f(0) = 0 \Rightarrow C = 0$

So, $f(1) = \frac{\pi}{4}(1) + \frac{(1)^2}{4}$

$f(1) = \frac{\pi+1}{4}$ Ans

Techniques of Integration

- Substitution
- Partial Fraction
- Miscellaneous
- By parts (Product rule)

Substitution :

Function ke arse part ko 't' assume karo jisko diff. karne pe pura f(x) hi 't' ki value mein convert hogaye.

Eg: $\int \frac{\cos(\ln x)}{x} \cdot dx$

Sol'n Let $\ln x = t \Rightarrow \frac{1}{x} \cdot dx = dt$

So, $f(x) = \int \cos t \cdot dt = \sin(t) + C$

$f(x) = \sin(\ln x) + C$ Ans

Eg: $\int \tan x \cdot dx$

Sol'n $f(x) = \int \frac{\sin x}{\cos x} \cdot dx$ | Let, $\cos x = t$, $\sin x \cdot dx = -dt$

$f(x) = \int \frac{-dt}{t} = -\ln t + C$

$f(x) = -\ln|\cos x| + C = \ln|\sec x| + C$

Eg: $\int \sec x \cdot dx$

Sol'n $f(x) = \int \frac{\sec x(\sec x + \tan x)}{(\sec x + \tan x)} \cdot dx$

$= \int \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} \cdot dx$

Let, $\sec x + \tan x = t \Rightarrow (\sec x \cdot \tan x + \sec^2 x)\,dx = dt$

So, $f(x) = \int \frac{dt}{t} = \ln(t) + C$

$f(x) = \ln(\sec x + \tan x) + C$

Important Formulas :→

1) $\int \tan x \cdot dx = \ln(\sec x) + C \;/\; -\ln(\cos x) + C$

2) $\int \cot x \cdot dx = \ln(\sin x) + C$

3) $\int \sec x \cdot dx = \ln(\sec x + \tan x) + C \;/\; \ln\tan\left(\frac{\pi}{2} + \frac{x}{2}\right)$

4) $\int \operatorname{cosec} x \cdot dx = \ln(\operatorname{cosec} x - \cot x) \;/\; \ln\left|\tan\frac{x}{2}\right| + C$

Ques $\int \frac{(2x-1)\cos\sqrt{(2x-1)^2+5}}{\sqrt{4x^2-4x+6}} \cdot dx$

Sol'n $4x^2 - 4x + 6 = 4x^2 - 4x + 1 + 4 + 1$

$= (2x-1)^2 + 5$

Let, $\sqrt{(2x-1)^2+5} = t$

$\frac{1}{\sqrt{(2x-1)^2+5}} \times \frac{1}{2x} \times 2(2x-1) \cdot 2 \cdot dx = dt$

So, $f(t) = \int \cot(t) \cdot \frac{dt}{2} \Rightarrow \frac{\sin(t)}{2} + C$

$f(x) = \frac{1}{2}\sin\sqrt{(2x-1)^2+5} + C$ Ans

Ques $\int \frac{e^{3\ln(2x)} + 5e^{2\ln(2x)}}{e^{4\ln(x)} + 5e^{3\ln(x)} - 7e^{2\ln(x)}} \cdot dx$

Sol'n $I = \int \frac{(2x)^3 + 5(2x)^2}{x^4 + 5(x)^3 - 7(x)^2} \cdot dx$

$I = \int \frac{8x^3 + 20x^2}{x^4 + 5x^3 - 7x^2} dx$

$I = \int \frac{4x^2(2x+5x)}{x^2(x^2+5x-7)} \cdot dx \Rightarrow 4\int \frac{2x+5}{x^2+5x-7} \cdot dx$

Let, $x^2 + 5x - 7 = t \Rightarrow (2x+5)\,dx = dt$

$I = 4\int \frac{dt}{t} \Rightarrow 4\ln(t) + C$

$I = 4\ln\left(\frac{x^2+5x-7}{x^2+5x-7}\right) + C$ Ans

Integrating-2

Ques $\int \frac{5x^8 + 7x^6}{(x^2+1+2x^7)^2} \cdot dx$

Whenever one bracket in denominator, take highest power out & convert all into -ve powers.

Sol'n $I = \int \frac{5x^8 + 7x^6}{x^{14}(x^{-5} + x^{-7} + 2)^2} \cdot dx$

$I = \int \frac{5x^{-6} + 7x^{-8}}{(x^{-5} + x^{-7} + 2)^2} \cdot dx$

Let $x^{-5} + x^{-7} + 2 = t$

$-5x^{-6} + (-7)x^{-8} = \frac{dt}{dx}$

$(5x^{-6} + 7x^{-8})\,dx = -dt$

$I = -\int \frac{dt}{t^2} = \frac{1}{t} + C$

$I = \frac{1}{x^5 + \frac{1}{x^7} + 2} + C \Rightarrow \frac{x^7}{x^9 + 2x^7 + 1} + C$ Ans

Ques If $\int \frac{\cos x\,dx}{\sin^3 x\,(1+\sin^6 x)^{2/3}} = f(x)(1+\sin^6 x)^{\lambda} + C$

then $\lambda f(\pi/3)$ is equal to ——.

Sol^n Let $\sin x = t \Rightarrow \cos x\,dx = dt$

$$I = \int \frac{dt}{t^3(1+t^6)^{2/3}} = \int \frac{dt}{t^3(t^6)^{2/3}(t^{-6}+1)^{2/3}}$$

$$I = \int \frac{t^{-7}}{(t^{-6}+1)^{2/3}}\cdot dt$$

Let, $t^{-6}+1 = z^3$

$-6t^{-7}\cdot dt = 3z^2\cdot dz$

$t^{-7}\cdot dt = \frac{z^2\cdot dz}{-2}$

$$I = \frac{1}{-2}\int \frac{z^2\,dz}{z^2} \Rightarrow \frac{-z}{2} + C$$

$$I = -\frac{1}{2}\frac{(1+\sin^6 x)^{1/3}}{\sin^2 x} + C$$

So, $f(x) = -\frac{1}{2\sin^2 x}$ and $\lambda = 3$

$$\lambda f(\pi/3) = 3\left(-\frac{1}{2}\times\frac{4}{3}\right) \Rightarrow \boxed{-2}$$ Ans

(*) Trending -2

Ques $\int \frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}}\cdot dx$

When two brackets with fractional power in denominator, multiply divide by (bracket with high power) raise to smaller power of either two.

Sol^n

$$I = \int \frac{1}{(x-1)^{3/4}(x+2)^{5/4}}\cdot dx$$

$$I = \int \frac{\frac{1}{(x+2)^{2}}}{\frac{(x-1)^{3/4}}{(x+2)^{3/4}}(x+2)^{5/4}}\,dx$$

Let, $\frac{x-1}{x+2} = t$

$\frac{1(x+2)-1(x-1)}{(x+2)^2}\cdot dx = dt$

$\frac{dx}{(x+2)^2} = \frac{dt}{3}$

$$I = \frac{1}{3}\int \frac{dt}{t^{3/4}}$$

$$I = \frac{1}{3}\cdot t^{1/4} + C \Rightarrow \boxed{\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + C}$$ Ans

Ques $\int \frac{dx}{(x+4)^{8/7}(x-3)^{6/7}}$

Sol^n

$$I = \int \frac{dx}{\left(\frac{x-3}{x+4}\right)^{6/7}(x+4)^{8/7+6/7}}$$

$$I = \int \frac{dx}{\left(\frac{x-3}{x+4}\right)^{6/7}(x+4)^2}$$

Let $\frac{x-3}{x+4} = t$

$\frac{1(x+4)-1(x-3)}{(x+4)^2} = \frac{dt}{dx}$

$\frac{dx}{(x+4)^2} = \frac{dt}{7}$

$$I = \frac{1}{7}\int \frac{dt}{t^{6/7}}$$

$$I = t^{1/7} + C = \boxed{\left(\frac{x-3}{x+4}\right)^{1/7} + C}$$ Ans

Ques $\int \frac{d\theta}{\cos^2\theta\,(\tan\theta + \sec 2\theta)}$

Sol^n

$$I = \int \frac{\sec^2\theta\cdot d\theta}{\tan 2\theta + \sec 2\theta}$$ $\left[\because \cos 2\theta = \frac{1+\tan^2\theta}{1-\tan^2\theta}\right]$

$$I = \int \frac{\sec^2\theta\,.\,d\theta}{\frac{2\tan\theta}{1-\tan^2\theta} + \frac{1}{\cos 2\theta}} = \int \frac{\sec^2\theta\cdot d\theta}{\frac{2\tan\theta}{1-\tan^2\theta} + \frac{1+\tan^2\theta}{1-\tan^2\theta}}$$

Let, $\tan\theta = t \Rightarrow \sec^2\theta\cdot d\theta = dt$

$$I = \int \frac{dt}{\left(\frac{1+2t+t^2}{1-t^2}\right)} = \int \frac{(1+t)(1-t)\,.\,dt}{(t+1)^2}$$

$$I = \int \frac{1-t}{1+t}\cdot dt \Rightarrow \int \frac{2-1-t}{1+t}\cdot dt$$

$$I = \int \frac{2}{1+t}\cdot dt - \int \frac{1+t}{1+t}\cdot dt$$

$$I = 2\ln|1+t| - t + C$$

$$I = 2\ln|1+\tan\theta| - \tan\theta + C$$ Ans

#General Substitution & Loving Integrals :

General Substitutions :→

1) $\sqrt{a^2-x^2}$; $x = a\sin\theta$
2) $\sqrt{a^2+x^2}$; $x = a\tan\theta$
3) $\sqrt{x^2-a^2}$; $x = a\sec\theta$
4) $\sqrt{\frac{a^2-x^2}{a^2+x^2}}$; $x^2 = a^2\cos 2\theta$

Ques $\int \frac{\sqrt{x}}{(1+x)^2}\,dx$

Sol^n Let, $x = \tan^2\theta \Rightarrow dx = 2\tan\theta\cdot\sec^2\theta\cdot d\theta$

$$I = \int \frac{\sqrt{\tan^2\theta}\,.\,2\tan\theta\cdot\sec^2\theta\cdot d\theta}{(1+\tan^2\theta)^2}$$

$$I = \int \frac{2\tan^2\theta\cdot\sec^2\theta}{\sec^4\theta}\,.\,d\theta \Rightarrow \int \frac{2\tan^2\theta}{\sec^2\theta}\,.\,d\theta$$

$$I = \int 2\sin^2\theta\cdot d\theta = \int (1-\cos 2\theta)\,.\,d\theta$$

$$I = \theta - \frac{\sin 2\theta}{2} + C \Rightarrow \tan^{-1}\sqrt{x} - \frac{2\sin\theta\cdot\cos\theta}{2} + C$$

$$I = \tan^{-1}\sqrt{x} - \frac{\sqrt{x}}{\sqrt{1+x^2}}\cdot\frac{1}{\sqrt{1+x}} + C$$

$$\Rightarrow \boxed{I = \tan^{-1}\sqrt{x} - \frac{\sqrt{x}}{1+x} + C}$$ Ans

Ques $\int \frac{dx}{(x^2-2x+10)^2}$

Sol^n

$$I = \int \frac{dx}{(x^2-2x+10+1-1)^2} = \int \frac{dx}{\left[(x-1)^2+9\right]^2}$$

Let, $x-1 = 3\tan\theta \Rightarrow dx = 3\sec^2\theta\cdot d\theta$

$$I = \int \frac{3\sec^2\theta\,.\,d\theta}{\left[9\tan^2\theta+9\right]^2} \Rightarrow \int \frac{3\sec^2\theta\,.\,d\theta}{(9\sec^2\theta)^2}$$

$$I = \int \frac{3}{81}\frac{\sec^2\theta}{\sec^4\theta}\,.\,d\theta \Rightarrow \frac{1}{27}\int \cos^2\theta\cdot d\theta$$

$$I = \int \frac{1+\cos 2\theta}{2\times 2}\,d\theta \Rightarrow \left(\theta + \frac{\sin 2\theta}{2}\right)\frac{1}{54} + C$$

$\therefore x-1 = 3\tan\theta \Rightarrow \tan\theta = \frac{x-1}{3}$

$\theta = \tan^{-1}\left(\frac{x-1}{3}\right)$

$\sin\theta = \frac{x-1}{\sqrt{(x-1)^2+3^2}}$; $\cos\theta = \frac{3}{\sqrt{(x-1)^2+3^2}}$

$$I = \frac{1}{54}\left(\tan^{-1}\left(\frac{x-1}{3}\right) + \frac{3(x-1)}{x^2-2x+10}\right) + C$$ Ans

Loving Integrals :→

1) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
2) $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
3) $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + C$
4) $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left(x+\sqrt{x^2-a^2}\right) + C$
5) $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x+\sqrt{x^2+a^2}\right) + C$

Solving Integrals using above formulas :→

1) $\frac{1}{\text{Quad.}}$ / $\frac{1}{\sqrt{\text{Quad.}}}$ → complete the square of given quad. and apply loving integrals

2) $\frac{\text{Linear}}{\text{Quad.}}$ / $\frac{\text{Linear}}{\sqrt{\text{Quad.}}}$ → Write linear eqn as, $px+q = M(\text{Quad})' + \lambda$ ⇓ Differentiation of Quad.

Eg: $\int \frac{5x+4}{\sqrt{x^2+2x+5}}\,dx$

$5x+4 = \lambda(2x+2) + M$

$5x+4 = 2\lambda x + (2\lambda+M)$

$\boxed{\lambda = 5/2}$ & $\boxed{M = -1}$

Sol^n

$$I = \int \frac{\frac{5}{2}(2x+2) + (-1)}{(x^2+2x+5)^{1/2}}\,dx$$

$$I = \frac{5}{2}\int \frac{2x+2}{\sqrt{x^2+2x+5}}\cdot dx - \int \frac{dx}{\sqrt{x^2+2x+5}}$$

Let, $x^2+2x+5 = t^2$

$(2x+2)dx = dt\cdot 2t$

$$I_1 = \frac{5}{2}\int \frac{2t\,dt}{(t)}$$

$$I_1 = 5\sqrt{x^2+2x+5}$$

$$I_2 = \int \frac{1}{\sqrt{x^2+2x+5+1-1}}\,dx$$

$$I_2 = \int \frac{1}{\sqrt{(x+1)^2-(2)^2}}\,dx$$

$$I_2 = \ln\left(x+1+\sqrt{x^2+2x+5}\right)$$

$$I = 5\sqrt{x^2+2x+5} - \ln\left(x+1+\sqrt{x^2+2x+5}\right) + C$$ Ans

#Integration by Parts :

$$\int (u\cdot v)\,dx = u\int v\,dx - \left(\frac{du}{dx}\int v\,dx\right)dx$$

1st 2nd

The order of functions while integrating by parts can be decided by,

I	L	A	T	E
Inv. T/g	Log	Algebraic	T/g	Exponential

Indefinite Integration - 2

(*) Type-01 : Separate functions are visible :→

Eg: $I = \int \underset{II}{x} \cdot \underset{I}{\tan^{-1}(x)} \cdot dx$

Soln

$I = \tan^{-1}x \int x \cdot dx - \int \frac{1}{x^2+1} \cdot \frac{x^2}{2} \cdot dx$

$I = \frac{x^2}{2} \cdot \tan^{-1}x - \frac{1}{2}\int \frac{x^2+1-1}{x^2+1} \cdot dx$

$I = \frac{x^2}{2} \cdot \tan^{-1}x - \frac{x}{2} + \frac{\tan^{-1}x}{2} + C$ Ans

(*) Type-02 : Considering '1' as the function :→

Eg: $I = \int \ln x \cdot$

Soln $I = \int \underset{II}{(1)} \underset{I}{(\ln x)} \cdot dx$

$I = \ln x \int 1 \cdot dx - \int \frac{1}{x} \cdot x \cdot dx$

$I = x\ln x - x + C$ Ans

NOTE: $\int \ln x \cdot dx = x \cdot \ln x - x + C$

(*) Type-03 : Recurring By Parts :→

Eg: $I = \int \underset{II}{e^x} \cdot \underset{I}{\cos x} \cdot dx$

Soln

$I = \cos x \cdot \int e^x \cdot dx - \int -\sin x \cdot e^x \, dx$ ((I) (II))

$I = e^x \cdot \cos x + \left[\sin x \cdot e^x - \int \cos x \cdot e^x \cdot dx\right]$ (↓ I)

$2I = e^x \cdot \cos x + e^x \cdot \sin x$

$\Rightarrow \boxed{I = \frac{e^x(\sin x + \cos x)}{2} + C}$ Ans

(*) Type-04 : special cases :→

1) $\boxed{\int e^x \left(f(x) + f'(x)\right) \cdot dx = e^x \cdot f(x) + C}$

2) $\boxed{\int \left(f(x) + x \cdot f'(x)\right) dx = x \cdot f(x) + C}$

Eg: $I = \int \frac{x \cdot e^x}{(1+x)^2} \cdot dx$

Soln

$I = \int e^x \cdot \left(\frac{x+1-1}{(1+x)^2}\right) dx \Rightarrow$

$I = \int e^x \left(\underset{f(x)}{\frac{1}{1+x}} + \underset{f'(x)}{\frac{-1}{(1+x)^2}}\right) dx$

$\boxed{I = e^x\left(\frac{1}{1+x}\right) + C}$ Ans

Eg: $I = \int \frac{x^2 e^x}{(x+2)^2} \cdot dx$

Soln

$I = e^x \int \frac{x^2-4+4}{(x+2)^2} \cdot dx$

$I = \int e^x \left(\underset{f(x)}{\frac{x-2}{x+2}} + \underset{f'(x)}{\frac{4}{(x+2)^2}}\right) \cdot dx$

$\boxed{I = e^x\left(\frac{x-2}{x+2}\right) + C}$ Ans

Eg: $I = \int \left(\sin(\ln x) + \underset{f'(x)}{\cos(\ln x)}\right) dx$ ($f(x)$)

Soln

$I = \int \left[\sin(\ln x) + \frac{x\cos(\ln x)}{x}\right] dx$

$\boxed{I = x \cdot \sin(\ln x) + C}$ Ans

Eg: $\int \left(2\ln x + (\ln x)^2\right) dx$

Soln $I = \int \underset{xf'(x)}{\frac{x(2\ln x)}{x}} + \underset{f(x)}{(\ln x)^2} \cdot dx$

$I = x \cdot \frac{(\ln x)^2}{2} + C$ Ans

Ques $\int e^x \cdot x^x (2 + \ln x) \cdot dx$

Soln $I = \int e^x \cdot x^x \left(1 + (1+\ln x)\right) dx$

$I = \int e^x \left(\underset{f(x)}{x^x} + \underset{f'(x)}{x^x(1+\ln x)}\right) dx$

$\boxed{I = e^x \cdot x^x + C}$ Ans

Ques $\int \left(1 + x - \frac{1}{x}\right) e^{x+\frac{1}{x}} \cdot dx$

Soln $I = \int \left(\underset{f(x)}{e^{x+\frac{1}{x}}} + \underset{xf'(x)}{x\left(1-\frac{1}{x^2}\right) \cdot e^{x+\frac{1}{x}}}\right) dx$

$\boxed{I = x \cdot e^{x+\frac{1}{x}} + C}$ Ans

Ques $I = \int \left(\sin^{-1}\sqrt{\frac{x}{1+x}}\right) dx$ (↓)

Soln Convert into ITF with simple expression.

$\because \sqrt{(\sqrt{x})^2 + \left((1+x)^{\frac{1}{2}}\right)^2} = \sqrt{1} = 1$

$I = \int \tan^{-1}\left(\frac{\sqrt{x}}{1}\right) \cdot dx$

$I = \int \underset{II}{(1)} \underset{I}{\tan^{-1}\sqrt{x}} \cdot dx$ (↓ ↓)

$I = \tan^{-1}\sqrt{x} \int dx - \int \frac{1}{1+x} \cdot x \cdot dx \cdot \frac{1}{2\sqrt{x}}$

$I = x \cdot \tan^{-1}\sqrt{x} - \frac{1}{2}\int \frac{\sqrt{x}}{1+x} \cdot dx$ (↓)

Let, $x = t^2$; $dx = 2t \cdot dt$; So, $I' = \int \frac{t \cdot 2t \cdot dt}{1+t^2}$

$I' = 2\int \frac{t^2+1-1}{1+t^2} \cdot dt$

$I' = 2\left(\int \frac{t^2+1}{t^2+1} \cdot dt - \int \frac{dt}{1+t^2}\right)$

$I' = 2\left[t - \tan^{-1}(t)\right]$

Hence,

$I = x\tan^{-1}\sqrt{x} - \sqrt{x} + \tan^{-1}\sqrt{x} + C$

$\boxed{I = (x+1)\tan^{-1}\sqrt{x} - \sqrt{x} + C}$ Ans

Ques $I = \int \left(\frac{x}{x\sin x + \cos x}\right)^2 \cdot dx$

Soln $I = \int \frac{x^2}{(x\sin x + \cos x)^2} \cdot dx$

Let, $x \cdot \sin x + \cos x = t$

$(x\cos x + \sin x - \sin x)dx = dt$

$\boxed{x\cos x \cdot dx = dt}$ (↓)

We have 'x' in numerator, so, we need to introduce $\cos x$ in num^r to make $f(x)$ integrable.

$I = \int \underbrace{\frac{x \cdot \cos x}{(x\sin x + \cos x)^2}}_{II} \cdot \underbrace{\frac{x}{\cos x}}_{I} \cdot dx$

(Integrable)

$I = \frac{x}{\cos x}\int \frac{x\cdot\cos x}{(x\sin x + \cos x)^2} \cdot dx - \int \left[\frac{1(\cos x) + x\cdot\sin x}{\cos^2 x} \times \int II \right] \cdot dx$

$I' = \int \frac{x\cdot\cos x}{(x\sin x + \cos x)^2} dx$

$= \int \frac{dt}{t^2} = \frac{-1}{t}$

$\left[I' = \frac{-1}{x\sin x + \cos x}\right]$

So,

$I = \frac{x}{\cos x}\left(\frac{-1}{x\sin x + \cos x}\right) - \int \frac{\cos x + x\sin x}{\cos^2 x} \cdot \frac{(-1)}{\cos x + x\sin x} \cdot dx$

$I = \frac{x}{\cos x}\left(\frac{-1}{x\sin x + \cos x}\right) + \int \sec^2 x \cdot dx$

$\boxed{I = \frac{-x\cdot\sec x}{x\sin x + \cos x} + \tan x + C}$ Ans

Shortcut for Integration by parts :→

Type-01 ⇒ $\int x^n \cdot f(x) \cdot dx$

1) Identify Ist func^n & IInd func^n
2) Differentiate Ist & integrate IInd until zero obtained
3) Assign alternate +/− sign and cross multiply from left to right only
4) Then simply add them.

Eg: $I = \int \underset{I}{x^2} \cdot \underset{II}{\cos 2x} \cdot dx$

Sol^n

	D	I
⊕	x^2	$\cos 2x$
⊖	$2x$	$\sin 2x/2$
⊕	2	$-\cos 2x/4$
⊖	0	$-\sin 2x/8$

$I = x^2\left(\frac{\sin 2x}{2}\right) + \frac{2x \cdot \cos 2x}{4} - \frac{2\sin 2x}{8} + C$ Ans

Eg: $I = \int \underset{I}{x^3} \cdot \underset{II}{e^{2x}} \cdot dx$

Sol^n

	D	I
⊕	x^3	e^{2x}
⊖	$3x^2$	$e^{2x}/2$
⊕	$6x$	$e^{2x}/4$
⊖	6	$e^{2x}/8$
⊕	0	$e^{2x}/16$

$I = \frac{x^3 \cdot e^{2x}}{2} - \frac{3x^2 \cdot e^{2x}}{4} + \frac{6x \cdot e^{2x}}{8} - \frac{6 \cdot e^{2x}}{16} + C$ Ans

Type-02 ⇒ $\int e^{ax} \cdot \cos(bx) \cdot dx$ (Recurring by parts)

Entire procedure is same as Type-01, the only difference is

① Here, stop at the point where we get same func^n (magnitude wise) while differentiating.

② At last, multiply end non-zero terms of (D & I) columns.

Eg: $\int \underset{II}{e^x} \cdot \underset{I}{\cos x} \cdot dx$

Sol^n

	D	I
⊕	$\cos x$	e^x
⊖	$-\sin x$	e^x
⊕	$-\cos x$	e^x

$I = e^x \cdot \cos x + e^x \cdot \sin x - \frac{e^x \cdot \cos x}{(-I)}$

$2I = e^x \cdot \cos x + e^x \cdot \sin x$

$I = \frac{e^x \cdot \cos x + e^x \cdot \sin x}{2}$ Ans

Eg: $\int e^{2x} \cdot \cos x \cdot dx$

Sol^n

	D	I
⊕	$\cos x$	e^{2x}
⊖	$-\sin x$	$e^{2x}/2$
⊕	$-\cos x$	$e^{2x}/4$

$I = \frac{e^{2x} \cdot \cos x}{2} + \frac{e^{2x} \cdot \sin x}{4} - \frac{I}{4}$

$\frac{5I}{4} = e^x\left(\frac{\sin x + \cos x}{4} \cdot \frac{1}{2}\right)$

$I = \frac{e^x(\sin x + 2\cos x)}{5}$ Ans

Integration by Partial Fraction :-

Concept Builders :-

1) $\int \frac{dx}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$

2) $\int \frac{dx}{(x+1)(x+2)(x+3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3}$

3) $\int \frac{dx}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$

4) $\int \frac{dx}{(1+x)^3(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} + \frac{D}{(x+2)}$

5) $\int \frac{dx}{(x^2+1)^2(x-1)} = \frac{Ax+B}{(x^2+1)} + \frac{Cx+D}{(x^2+1)^2} + \frac{E}{(x-1)}$

Eg: $\int \frac{3x-2}{(x+1)^2(x+3)} \cdot dx$

Sol^n $\frac{3x-2}{(x+1)^2(x+3)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+3}$

$3x-2 = A(x+1)(x+3) + B(x+3) + C(x+1)^2$

$x=-1$; $-5 = 2B \Rightarrow \boxed{B = -5/2}$

$x=-3$; $-11 = 4C \Rightarrow \boxed{C = -11/4}$

$x=0$; $-2 = 3A + 3B + C$

$\Rightarrow \boxed{A = 11/4}$

So,

$I = -\frac{5}{2}\int \frac{1}{(x+1)^2} \cdot dx - \frac{11}{4}\int \frac{1}{x+3} \cdot dx + \frac{11}{4}\int \frac{1}{x+1} \cdot dx$

$I = \frac{5}{2(x+1)} - \frac{11}{4}\ln|x+3| + \frac{11}{4}\ln|x+1| + C$ Ans

Important Formulas derived by Partial fraction :-

1) $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\ln\left(\frac{a+x}{a-x}\right) + C$

2) $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right) + C$

PROOF ⇒ $I = \int \frac{1}{x^2 - a^2} \cdot dx$

$I = \frac{1}{2a}\int \frac{(x+a)-(x-a)}{(x-a)(x+a)} \cdot dx = \frac{1}{2a}\int \frac{1}{x-a} - \frac{1}{x+a} \cdot dx$

$I = \frac{1}{2a}\left(\ln(x-a) - \ln(x+a)\right) + C$

$I = \frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right) + C$

Ques $\int \frac{\cos\theta}{5 + 7\sin\theta - 2\cos^2\theta} \cdot d\theta$

Sol^n Let, $\sin\theta = t \Rightarrow \cos\theta \cdot d\theta = dt$

$I = \int \frac{dt}{5 + 7t - 2(1-t^2)}$

$I = \int \frac{dt}{2t^2 + 7t + 3} \Rightarrow \int \frac{dt}{2t^2 + \frac{7t}{1} + \frac{3}{2} + \left(\frac{7}{4}\right)^2 - \left(\frac{7}{4}\right)^2}$

$I = \int \frac{dt}{\left(2x + \frac{7}{2}\right)^2 - \left(\frac{\sqrt{37}}{2}\right)^2}$ (For applying completing sq. method, always make (coeff. of x^2) = 1)

$I = \int \frac{dt}{\left(t + \frac{7}{4}\right)^2 - \left(\frac{5}{4}\right)^2}$

$I = \frac{2}{5}\ln\left(\frac{\sin\theta + 1/2}{\sin\theta + 3}\right) + C$ Ans

NOTE: 1) Partial fraction can be applied only if [Degree (Num^r) < Degree (Den^r)]

2) If [Degree (Num^r) ≥ Degree (Den^r)] then we use 'long Division.'

Eg: $\int \frac{x^2+1}{x^2-5x+6} \cdot dx$ (To avoid using long division, try to make denominator expression in numerator.)

Sol^n $I = \int \frac{(x^2-5x+6) + (1+5x-6)}{x^2-5x+6} \cdot dx$

$I = \int 1 \cdot dx + \int \frac{5x-5}{(x-2)(x-3)} \cdot dx$

$5x-5 = A(x-3) + B(x-2)$

$x=3$; $\boxed{10 = B}$ | $x=2$; $\boxed{5 = -A}$

$\boxed{I = x - 5\ln(x-2) + 10\ln(x-3) + C}$ Ans

* Partial fraction Involving even powers of x only :-

Eg: $\int \frac{x^2+1}{(x^2+2)(x^2+3)} \cdot dx$ (Temporarily, substitute x^2 as t.)

Sol^n $\frac{t+1}{(t+2)(t+3)} = \frac{A}{(t+2)} + \frac{B}{(t+3)}$

$t+1 = A(t+3) + B(t+2)$

$x=-3$; $\boxed{B = 2}$ | $x=-2$; $\boxed{A = -1}$

$I = -\int \frac{1}{x^2+2} \cdot dx + 2\int \frac{1}{x^2+3} \cdot dx$

$I = -\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{2}{\sqrt{3}} \cdot \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$ Ans

Integration of Trigonometric Functions :-

Type-01: Quadratic in sin and cos in denominator

$\int \frac{dx}{a + b\sin^2 x}$ / $\int \frac{dx}{a + b\cos^2 x}$ / $\int \frac{dx}{a\sin^2 x + b\cos^2 x + c\sin x \cdot \cos x}$ / $\int \frac{dx}{(a\cos x + b\sin x)^2}$

(Multiply numerator and denominator by $\sec^2 x$ or $\operatorname{cosec}^2 x$, and then assume $\tan x = t$.)

Eg: $\int \frac{dx}{4 - 5\sin^2 x} \times \frac{\sec^2 x}{\sec^2 x}$

Sol^n $I = \int \frac{\sec^2 x \cdot dx}{4\sec^2 x - 5\tan^2 x} \Rightarrow \int \frac{\sec^2 x \cdot dx}{4(1 + \tan^2 x) - 5\tan^2 x}$

$\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$

$I = \int \frac{dt}{4(1+t^2) - 5t^2} \Rightarrow \int \frac{dt}{4 - t^2}$

$I = \frac{1}{4}\ln\left(\frac{2+t}{2-t}\right) + C$

$\boxed{I = \frac{1}{4}\ln\left(\frac{2 + \tan x}{2 - \tan x}\right) + C}$ Ans

Type-02: Linear in sin or cos or both in denominator

$\int \frac{dx}{a + b\sin x}$ / $\int \frac{dx}{a + b\cos x}$ / $\int \frac{dx}{a + b\sin x + c\cos x}$

Indefinite Integration-3

Convert $\sin x$ & $\cos x$ into their corresponding $\tan(x/2)$ form and assume $\tan(\frac{x}{2}) = t$.

$$\sin x = \frac{2\tan(x/2)}{1+\tan^2(x/2)} \qquad \cos x = \frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}$$

Shortcut ⇒ We can directly substitute T-ratios in terms of t as →

$dx \to \frac{2\,dt}{1+t^2}$; $K \to K(1+t^2)$; $\cos x \to \frac{1-t^2}{1+t^2}$; $\sin x \to \frac{2t}{1+t^2}$; where $t = \tan\frac{x}{2}$

Eg: $\int \frac{dx}{5+4\cos x}$

Sol^n

$$I = \int \frac{dx}{5+4\left[\frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}\right]}$$

$$I = \int \frac{dx\,(1+\tan^2(x/2))}{5(1+\tan^2(x/2)) + 4(1-\tan^2(x/2))} \qquad (\sec^2(\tfrac{x}{2}))$$

Let, $t = \tan(\frac{x}{2}) \Rightarrow dt = \sec^2(\frac{x}{2}) \times \frac{1}{2}$

$$I = \int \frac{2\cdot dt}{5(1+t^2)+4(1-t^2)} \Rightarrow \int \frac{2dt}{t^2+9}$$

$$I = \frac{2}{3}\tan^{-1}\left(\frac{t}{3}\right)+C \Rightarrow \frac{2}{3}\tan^{-1}\left(\frac{\tan\frac{x}{2}}{3}\right)+C \quad \text{Ans}$$

Eg: $\int \frac{dx}{\sin x - 3\cos x - 1}$

Sol^n Using Shortcut

$$I = \int \frac{2dt}{2t-3(1-t^2)-1(1+t^2)}$$

$$I = \int \frac{2\cdot dt}{2t^2+2t-4} \Rightarrow \int \frac{dt}{t^2+t-2+\frac{1}{4}-\frac{1}{4}}$$

$$I = \int \frac{dt}{\left(t+\frac{1}{2}\right)^2 - \frac{9}{4}} \Rightarrow \frac{1}{2(\frac{3}{2})}\ln\left(\frac{t+\frac{1}{2}-\frac{3}{2}}{t+\frac{1}{2}+\frac{3}{2}}\right)+C$$

$$\boxed{I = \frac{1}{3}\ln\left(\frac{\tan\frac{x}{2}-1}{\tan\frac{x}{2}+2}\right)+C} \quad \text{Ans}$$

*** Type-03 :** Linear expressions of $\sin$ & $\cos$ present in both Numr & Denr.

$$\int \frac{a\sin x + b\cos x + c}{p\sin x + q\cos x + r}\,dx \qquad N^r = A(D^r) + B\left(\frac{d}{dx}D^r\right) + C$$

Write the numerator in this form where A, B & C are constants.

Eg: $\int \frac{\sin x + 3\cos x + 1}{\sin x - 3\cos x - 1}\,dx$

Sol^n $\sin x + 3\cos x + 1 = A(\sin x - 3\cos x - 1) + B(\cos x + 3\sin x) + C$

$A + 3B = 1 \;]\times 3$; $-3A + B = 3$ (add) $\Rightarrow 10B = 6 \Rightarrow \boxed{B = 3/5}$

$\boxed{A = -4/5}$

$-A + C = 1 \Rightarrow \boxed{C = 1/5}$

$$I = \int \frac{-\frac{4}{5}(\sin x - 3\cos x - 1) + \frac{3}{5}(\cos x + 3\sin x) + \frac{1}{5}}{\sin x - 3\cos x - 1}\,dx$$

$$I = -\frac{4}{5}\int dx + \frac{3}{5}\int \frac{\cos x + 3\sin x}{\sin x - 3\cos x - 1}\,dx + \frac{1}{5}\int \frac{dx}{\sin x - 3\cos x - 1}$$

Let, $\sin x - 3\cos x - 1 = t$; $(\cos x + 3\sin x)dx = dt$; $I' = \frac{3}{5}\int \frac{dt}{t} = \ln(t)^{3/5}$

Solve using type-2

$$I = -\frac{4x}{5} + \ln(\sin x - 3\cos x - 1)^{3/5} + \frac{1}{5}\left(\frac{1}{3}\ln\left(\frac{\tan\frac{x}{2}-1}{\tan\frac{x}{2}+1}\right)\right)+C \quad \text{Ans}$$

*** Type-04 :** Functions having $(x^2 \pm 1)$ in N^r and biquadratic in terms of x in D^r,

$$\int \frac{x^2+1}{x^4+Kx^2+1}\,dx \Big/ \int \frac{x^2-1}{x^4+Kx^2+1}\,dx$$

↳ Divide N^r & D^r by x^2 and take suitable substitution such that

① Upar (+) wala ho toh neeche minus wala bane

$$1 + \frac{1}{x^2} \longrightarrow x - \frac{1}{x}$$

② Upar (-) wala ho toh neeche plus wala bane

$$1 - \frac{1}{x^2} \longrightarrow x + \frac{1}{x}$$

Ques $\int \frac{x^2+1}{x^4+7x^2+1}\,dx$

Sol^n

$$I = \int \frac{x^2+1}{x^4+7x^2+1}\,dx \Rightarrow \int \frac{1+1/x^2}{x^2+7+1/x^2}\,dx$$

$$I = \int \frac{(1+1/x^2)}{x^2+\frac{1}{x^2}-2+2+7}\,dx \Rightarrow \int \frac{(1+1/x^2)}{\left(x-\frac{1}{x}\right)^2+9}\,dx$$

Let, $x - \frac{1}{x} = t \Rightarrow \left(1+\frac{1}{x^2}\right)dx = dt$

So,

$$I = \int \frac{dt}{t^2+9} = \frac{1}{3}\tan^{-1}\left(\frac{t}{3}\right)+C$$

$$\boxed{I = \frac{1}{3}\tan^{-1}\left(\frac{x-1/x}{3}\right)+C} \quad \text{Ans}$$

Ques $\int \frac{x^2-1}{x^4+x^2+1}\,dx$

Sol^n

$$I = \int \frac{(1-1/x^2)}{x^2+1+1/x^2}\,dx$$

Let, $x + \frac{1}{x} = t$; $\left(1-\frac{1}{x^2}\right)dx = dt$

$$I = \int \frac{(1-1/x^2)}{(x+\frac{1}{x})^2-1}\,dx$$

$$I = \int \frac{dt}{t^2-1} = \frac{1}{2(1)}\ln\left(\frac{t-1}{t+1}\right)+C$$

$$\boxed{I = \frac{1}{2}\ln\left(\frac{x^2-x+1}{x^2+x+1}\right)+C} \quad \text{Ans}$$

Type-05 : When $\sin x$ & $\cos x$ are in product with odd-even powers

$$\int \sin^m x \cdot \cos^n x\,dx$$

- Powers are (even-odd) or (odd-even) ⇒ Substitute the T-ratio having even powers
- Powers are fractional, -ve, both odd, both even or any other case ⇒ Make $\tan x$ and substitute it to 't'.

Eg: $\int \sin^5 x \cdot \cos^3 x\,dx$

Sol^n Let, $\cos x = t \Rightarrow -\sin x\,dx = dt$

$$I = -\int \sin^3 x \cdot t^4 \cdot dt \Rightarrow -\int (1-\cos^2 x)^2 \cdot t^4 \cdot dt$$

$$I = -\int (1-t^2)^2 \cdot t^4 \cdot dt \Rightarrow \int (-t^4 - t^8 + 2t^6)\,dt$$

$$I = -\frac{t^5}{5} - \frac{t^9}{9} + \frac{2t^7}{7} + C$$

$$\Rightarrow \boxed{I = \frac{-\cos^5 x}{5} - \frac{\cos^9 x}{9} + \frac{2\cos^7 x}{7} + C} \quad \text{Ans}$$

Eg: $\int \sin^{2/3} x \cdot \cos^{-8/3} x \cdot dx$

Sol^n

$$I = \int \left(\frac{\sin x}{\cos x}\right)^{\frac{2}{3}} \cdot (\cos x)^{-\frac{8}{3}+\frac{2}{3}} \cdot dx$$

$$I = \int (\tan x)^{2/3} \cdot \sec^2 x \cdot dx$$

Let, $\tan x = t \Rightarrow \sec^2 x \cdot dx = dt$

$$I = \int t^{2/3} \cdot dt \Rightarrow 3\frac{t^{5/3}}{5} + C$$

$$\boxed{I = 3\frac{(\tan x)^{5/3}}{5}} \quad \text{Ans}$$

Ques $\int \sec^{2/3} x \cdot \operatorname{cosec}^{4/3} x \cdot dx$

Sol^n

$$I = \int \frac{1}{(\cos x)^{2/3}} \cdot \frac{1}{(\sin x)^{4/3}} \cdot dx$$

$$I = \int \frac{dx}{(\cos x)^{\frac{2}{3}+\frac{4}{3}} \cdot \left(\frac{\sin x}{\cos x}\right)^{4/3}}$$

$$I = \int \frac{\sec^2 x \cdot dx}{(\tan x)^{4/3}}$$

Let, $\tan x = t$; $\sec^2 x \cdot dx = dt$

$$I = \int \frac{dt}{t^{4/3}} \Rightarrow \frac{-3}{t^{1/3}} + C \Rightarrow \frac{-3}{\sqrt[3]{\tan x}} + C \quad \text{Ans}$$

Type-06 : Sum of sin & cos in N^r and product of sin & cos in D^r.

↳ In such integrals, assume the conjugate of N^r $(\sin x \pm \cos x)$ as t, t^{-1}.

Ques $\int (\sqrt{\tan x} + \sqrt{\cot x})\,dx$

Sol^n $I = \int \left(\sqrt{\frac{\sin x}{\cos x}} + \sqrt{\frac{\cos x}{\sin x}}\right) dx$

$I = \int \frac{\sin^2 x + \cos^2 x}{\sqrt{\sin x \cdot \cos x}} \cdot dx$

Let, $\sin x - \cos x = t \Rightarrow (\cos x + \sin x)\,dx = dt$

$(\sin x - \cos x)^2 = t^2$

$\sin^2 x + \cos^2 x - 2\sin x \cdot \cos x = t^2$

$\Rightarrow \left[\sin x \cdot \cos x = \frac{t^2 - 1}{-2}\right]$

So,

$I = \int \frac{\sqrt{2}\,dt}{\sqrt{1 - t^2}} \Rightarrow \sqrt{2} \cdot \sin^{-1}(t) + C$

$\boxed{I = \sqrt{2}\sin^{-1}(\sin x - \cos x) + C}$ Ans

Ques $\int \frac{(\cos x - \sin x)\,dx}{\sqrt{8 - \sin 2x}}$

Sol^n Let, $\sin x + \cos x = t$

$(\cos x - \sin x)\,dx = dt$

$(\sin x + \cos x)^2 = t^2$

$1 + \sin 2x = t^2 \Rightarrow [\sin 2x = t^2 - 1]$

$I = \int \frac{dt}{\sqrt{8 - t^2 + 1}} = \int \frac{dt}{\sqrt{9 - t^2}}$

$I = \sin^{-1}\left(\frac{t}{3}\right) + C \Rightarrow \sin^{-1}\left(\frac{\sin x + \cos x}{3}\right) + C$ Ans

Integration of $\sqrt{\text{Quadratic}}$:

1) $\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log\left|x + \sqrt{x^2 - a^2}\right| + C$

2) $\int \sqrt{x^2 + a^2}\,dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\log\left|x + \sqrt{x^2 + a^2}\right| + C$

3) $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$

Proof :→ $\int \sqrt{x^2 - a^2}\,dx$

$I = \int_{II}^{I} \sqrt{x^2 - a^2}\,dx = \sqrt{x^2 - a^2} \cdot x - \int \frac{1}{2\sqrt{x^2 - a^2}} \cdot 2x^2\,dx$

$I = x\sqrt{x^2 - a^2} - \int \frac{x^2 - a^2 + a^2}{\sqrt{x^2 - a^2}}\,dx$

$I = x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2}\,dx - \int \frac{a^2}{\sqrt{x^2 - a^2}}\,dx$

$2I = x\sqrt{x^2 - a^2} - a^2 \ln\left(x + \sqrt{x^2 - a^2}\right) + C$

$I = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln\left|x + \sqrt{x^2 - a^2}\right| + C$

Eg : $\int \sqrt{x^2 - 8x + 7}\,dx$

Sol^n $I = \int \sqrt{(x-4)^2 - 3^2}\,dx$

$I = \frac{x-4}{2}\sqrt{(x-4)^2 - 9} - \frac{9}{2}\ln\left(x - 4 + \sqrt{(x-4)^2 - 9}\right) + C$ Ans

Integration of Irrational Functions :

- $\frac{1}{L\sqrt{L_0}}$: Root ke andar wale linear ko t^2 put karo. i.e. $\boxed{L_0 = t^2}$
- $\frac{1}{Q\sqrt{L_0}}$: Root ke andar wale linear ko t^2 put karo.
- $\frac{1}{L\sqrt{Q}}$: Linear ko $1/t$ put karo. $\boxed{L = \frac{1}{t}}$
- $\frac{1}{Q\sqrt{Q}}$: When Q outside root is factorisable → Factorise outside Q and apply 'Partial fraction'. Then Let linear factors as $1/t$.
- When Q outside root is perfect sq. → Convert outside into perfect sq. $(lx+m)^2$ and put $lx+m = 1/t$
- When both Q having missing $(x)^1$ term → Put, $x = \frac{1}{t}$.

Eg : $\int \frac{dx}{(x+2)\sqrt{x+1}} \Rightarrow \frac{1}{L\sqrt{L_0}}$

Sol^n Let, $x + 1 = t^2 \Rightarrow 2t \cdot dt = 1 \cdot dx$

$I = 2\int \frac{t \cdot dt}{(t^2 - 1 + 2) \cdot t} \Rightarrow 2\int \frac{dt}{t^2 + 1}$

$I = 2\tan^{-1}(t) + C \Rightarrow \boxed{2\tan^{-1}\left(\sqrt{x+1}\right) + C}$ Ans

Eg : $\int \frac{dx}{(x^2 + 5x + 6)\sqrt{x+1}} \Rightarrow \frac{1}{Q\sqrt{L_0}}$

Sol^n Let, $x + 1 = t^2 \Rightarrow 1 \cdot dx = 2t \cdot dt$

$I = \int \frac{2t \cdot dt}{\left[(t^2 - 1)^2 + 5(t^2 - 1) + 6\right] t}$

$I = \int \frac{2\,dt}{t^4 + 3t^2 + 2} \Rightarrow \int \frac{2 \cdot dt}{(t^2 + 1)(t^2 + 2)}$

$I = 2\int \frac{(t^2 + 2) - (t^2 + 1)}{(t^2 + 1)(t^2 + 2)} \Rightarrow 2\int \frac{dt}{t^2 + 1} - 2\int \frac{dt}{t^2 + 2}$

$I = 2\tan^{-1}(t) - \frac{2}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right) + C$

$I = 2\tan^{-1}\left(\sqrt{x+1}\right) - \sqrt{2}\tan^{-1}\left(\frac{\sqrt{x+1}}{\sqrt{2}}\right) + C$ Ans

Eg : $\int \frac{dx}{(x+1)\sqrt{1 + x - x^2}} \Rightarrow \frac{1}{L\sqrt{Q}}$

Sol^n Let, $x + 1 = \frac{1}{t} \Rightarrow 1 \cdot dx = \frac{-dt}{t^2}$

$I = -\int \frac{dt}{t^2 \cdot (1/t)\left[1 + \left(\frac{1}{t} - 1\right) - \left(\frac{1}{t} - 1\right)^2\right]^{1/2}}$

$I = \int \frac{-dt}{t^2 \left(\frac{1}{t}\right)\left(\frac{1}{t}\right)\sqrt{3t^2 - 1 - t^2}} \Rightarrow -\int \frac{dt}{\sqrt{3t^2 - 1 - t^2}}$

$I = \int \frac{-dt}{\sqrt{-t^2 + 3t - \frac{9}{4} + \frac{9}{4} - 1}^{1/2}}$

$I = \int \frac{-dt}{\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(t - \frac{3}{2}\right)^2}} \Rightarrow$ [illegible] $\sin^{-1}\left(\frac{t - \frac{3}{2}}{\frac{\sqrt{5}}{2}}\right) + C$

$\boxed{I = \sin^{-1}\left(\frac{2}{\sqrt{5}}\left(\frac{x+1}{2} - 3\right)\right) + C}$ Ans

Eg : $\int \frac{dx}{(x^2 + 4)\sqrt{4x^2 + 1}} \Rightarrow \frac{1}{Q\sqrt{Q}}$ without any term of $(x)^1$

Sol^n $x = \frac{1}{t} \Rightarrow 1 \cdot dx = \frac{-1}{t^2} \cdot dt$

$I = \int \frac{-dt/t^2}{\left(\frac{1}{t^2} + 4\right)\sqrt{\frac{4}{t^2} + 1}} \Rightarrow \frac{-dt/t^2}{\frac{(4t^2 + 1)}{t^2}\sqrt{\frac{t^2 + 4}{t^2}}}$

$I = \int \frac{-t \cdot dt}{4t^2 + 1\sqrt{t^2 + 4}}$ | Let, $4 + t^2 = z^2$, $2t \cdot dt = 2z \cdot dz$, $t \cdot dt = z \cdot dz$

$I = \int \frac{-z \cdot dz}{\left[4(z^2 - 4) + 1\right] z} \Rightarrow -\int \frac{dz}{4z^2 - 15}$

$I = -\frac{1}{4}\int \frac{dz}{z^2 - \left(\frac{\sqrt{15}}{2}\right)^2} \Rightarrow -\frac{1}{4} \cdot \frac{1}{2\left(\frac{\sqrt{15}}{2}\right)}\ln\left(\frac{z - \frac{\sqrt{15}}{2}}{z + \frac{\sqrt{15}}{2}}\right) + C$

$I = \ln\left(\frac{2\sqrt{4 + \frac{1}{x^2}} - \sqrt{15}}{2\sqrt{4 + \frac{1}{x^2}} + \sqrt{15}}\right) \cdot \left(\frac{-1}{4\sqrt{15}}\right) + C$ Ans

Eg : $\int \frac{dx}{(x^2 - x - 2)\sqrt{x^2 + x + 1}} \Rightarrow \frac{1}{Q\sqrt{Q}}$ with factorisable Q

Sol^n $I = \int \frac{dx}{\sqrt{x^2 + x + 1}} \times \left(\frac{A}{(x-2)} + \frac{B}{(x+1)}\right)$

$I = A\int \frac{dx}{(x-2)\sqrt{x^2 + x + 1}} + B\int \frac{dx}{(x+1)\sqrt{x^2 + x + 1}}$

Put $L = 1/t$; Put, $L = 1/t$